



COMPREHENSIVE USER GUIDE

CorpusMind

Local-first, AI-native research environment
for corpus linguistics and multimodal discourse analysis

ENGINE

Python + FastAPI + spaCy

PLATFORMS

Win / Mac / Linux / PWA

LICENSE

AGPL-3.0

v0.1.0

Dr. Waleed Mandour

Sultan Qaboos University | ORCID: 0000-0002-9262-5993

Prof. Wesam Ibrahim

Princess Nourah Bint Abdulrahman University | ORCID: 0000-0003-0710-6038

PART 1

Citation

If you use CorpusMind in your research, please cite it as:

Mandour, W., & Ibrahim, W. (2026). **CorpusMind: A local-first, AI-native research environment for corpus linguistics and multimodal discourse analysis** (Version 0.1.0) [Computer software]. Zenodo. <https://doi.org/10.5281/zenodo.21226650>

PART 2

1. What is CorpusMind?

CorpusMind is a research tool for corpus linguists and discourse analysts. It lets you upload texts, run concordance searches, compute collocations and keyness, analyze grammar and discourse features, and work with Arabic corpora using CAMEL Tools. It also includes a Vision suite for multimodal discourse analysis of images using Kress and van Leeuwen's Visual Grammar.

The AI Assistant answers your questions about the corpus by calling the analysis tools and citing specific evidence (concordance line IDs, computed statistics). Every answer is either "grounded" (backed by a tool call) or clearly flagged as "ungrounded."

Everything runs on your own machine. No data leaves your computer unless you explicitly choose to use a cloud AI provider.

PART 3

2. Installation

You need three things:

1. **Python 3.12 or newer** from python.org
2. **Node.js 20 or newer** from nodejs.org
3. **Ollama** from ollama.com (for the AI Assistant to work locally)

After installing Ollama, open a terminal and run:

```
ollama pull llama3.2:3b
```

Set Up the Engine

```
cd CorpusMind/engine python3 -m venv .venv source .venv/bin/activate pip
install -e ".[dev]" python -m spacy download en_core_web_sm corpusmind-engine
```

Set Up the Web App

Open a new terminal:

```
cd CorpusMind/web npm install npm run dev
```

Open <http://localhost:5173> in your browser.

Optional: Arabic Support

```
pip install camel-tools pyrsistent muddler cachetools emoji future regex  
camel_data -i morphology-db-msa-r13 camel_data -i dialectid-model6
```

PART 4

3. Creating a Project and Uploading Texts

1. Click **Projects** in the sidebar.
2. Click + **New** to create a project. Enter a name and select a language.
3. Click your project to select it.
4. Click + **New** under Corpora to create a corpus.
5. Drag and drop text files into the upload area.

Supported formats: TXT, DOCX, PDF, HTML, XML, CSV, Markdown.

The engine automatically detects encoding, cleans the text, tokenizes, tags with POS, lemmatizes, and parses dependencies using spaCy. The pipeline recipe (model name and version) is visible and exportable for reproducibility.

PART 5

4. Concordance Search (KWIC)

1. Select a corpus.
2. Click **Concordance** in the sidebar.
3. Type a search query.
4. Choose the level: word, lemma, or POS tag.
5. Set the context window (how many words to show left and right).
6. Click **Search**.

Each result line has a stable ID (for example, `doc:0:3`) that the AI Assistant can cite as evidence. Use `*` for wildcards.

Click **Export Excel** to download results.

PART 6

5. Frequency, Collocation, Keyness, and Dispersion

Frequency: Click any analysis tool in the sidebar. Choose word, lemma, or POS. The table shows frequency, per-million, and percent. STTR (Standardized Type-Token Ratio) is displayed at the top.

Collocation: Type a node word, set the window size (default 5 tokens each side), and click Compute. All 7 measures are shown: MI, T-score, log-likelihood, Dice, LogDice, chi-square, and Delta P (both directions).

Keyness: Select your target corpus, then choose a reference corpus. Results show both significance tests (log-likelihood, chi-square) and effect-size measures (Log Ratio, %DIFF, Simple Maths, Odds Ratio) side by side. Click **Methods PDF** to auto-draft a methodology paragraph for your manuscript.

Dispersion: Type a term. Results show Juilland's D (0 to 1, higher is more even) and Gries' DP (0 to $n-1/n$, lower is more even) across documents.

PART 7

6. Advanced Analysis Tools

N-grams: Choose n (2 to 10). Set minimum frequency and minimum range (number of distinct documents the n-gram must appear in). This follows the Biber et al. (1999) frequency-and-range criterion for lexical bundles.

POS Analysis: View POS distribution or POS n-grams (bigrams, trigrams, etc.) for stylistic analysis.

Grammar: Detect passive voice, modals, negation, relative clauses, complex noun phrases, and tense from the dependency parse.

Dependency: Query specific relations (nsubj, obj, iobj, obl, etc.) to find common governor-dependent pairs.

Discourse: Run Hyland's (2005) metadiscourse taxonomy across your corpus: transitions, frame markers, hedges, boosters, attitude markers, self-mentions, engagement markers, and more.

Vocabulary: Profile your corpus into frequency bands (K1, K2-K9, AWL, Off-list). Identify rare words and academic words.

Sentiment: Per-sentence sentiment scores (-1 to +1) with a timeline visualization.

Metaphor: Generates metaphor candidates (verbs with abstract subjects). These are candidates only. The LLM triages them, and a human must verify before any candidate counts as a confirmed metaphor.

PART 8

7. Arabic Analysis

Click **Arabic Tools** in the sidebar. Available tools:

- Morphology:** Full analysis with root (al-jizr), pattern (al-wazn), lemma, POS, stem, Buckwalter transliteration, number, gender, and broken-plural flag. Uses CAMEL Tools (calima-msa-r13).
 - Roots:** Extracts triliteral roots. For example, all words sharing the root k.t.b: kitab, maktaba, katib, yaktub.
 - Dialect ID:** Identifies MSA, Egyptian, Gulf, or Levantine using the full CAMEL DIDModel6 (6 city dialects).
 - Buckwalter:** Transliterates Arabic script to ASCII Latin.
 - Dediacritize:** Removes harakat (diacritics).
 - Normalize:** Unifies alef variants, teh marbuta, alef maksura.
 - Register:** Detects Classical, MSA, or Dialectal.
 - Translate:** Looks up Arabic-to-English translation equivalents.
-

PART 9

8. Vision Suite (Multimodal Analysis)

Click **Vision Suite** in the sidebar.

1. Create an image set within your corpus.
2. Drag and drop images (JPG, PNG, TIFF, WebP).
3. Optionally add captions.
4. Select an image and use the tabs:

Analyse: Colour analysis (dominant colours, warm/cold balance, brightness, contrast, saturation, symbolism notes), composition analysis (information value: Given/New, Ideal/Real, centre/margin, salience, rule of thirds, visual balance, framing), and OCR text extraction.

Visual Grammar: Analyses the image against Kress and van Leeuwen's (2006) three metafunctions: Representational, Interactive, and Compositional. Every claim is phrased as a hypothesis: "Under a Kress and van Leeuwen reading, X may indicate Y."

Align: Multimodal image-text alignment. Type the co-occurring text and click Align. The engine extracts image regions and text spans, then matches them with confidence scores. Cross-modal relations (reinforcement, complementarity, silence) are detected.

Discourse: 8 framework-lensed analyses: Social Semiotic, CDA (4 frameworks: Fairclough, van Dijk, Wodak, Machin and Mayr), Persuasion (Aristotle + Toulmin), Framing (Entman), Narrative (Labov), Visual Metaphor, Combined Emotion, and Cultural analysis.

PART 10

9. The AI Assistant

Click **AI Assistant** in the sidebar. The Assistant is a tool-using agent, not a chatbot. When you ask a question, it selects and calls the appropriate analysis tool, then writes its answer based on the tool's output.

If the Assistant calls a tool, the answer is marked **grounded** (green badge). If no tool was called, the answer is marked **ungrounded** (orange badge). The UI never silently presents an ungrounded answer as fact.

Example questions:

- bullet "What are the top 10 most frequent words in this corpus?"
 - bullet "Find all occurrences of 'research' and show me their contexts."
 - bullet "What are the strongest collocates of 'dog' within 5 tokens?"
 - bullet "Compare this corpus against the reference. What are the top keywords?"
 - bullet "What hedges does this author use?"
-

PART 11

10. Export, Collaboration, and Privacy

Export: Excel for concordance, frequency, collocations, and keyness. PDF for the auto-drafted Methods Section (cites exact tools, versions, and formulas).

Saved Searches and Bookmarks: Save queries with parameters for re-use. Bookmark specific concordance lines or statistics with notes.

Project Sharing: Mark a project as shared (public or private) with a share token. Sync events are logged in an audit trail.

At-Rest Encryption: Optional AES-256-GCM encryption for image files on disk. Enable by setting the `CORPUSMIND_ENCRYPTION_KEY` environment variable. The key is never stored on disk.

Accessibility: WCAG 2.1 AA target. Visible focus indicators, skip-to-content link, high contrast mode, reduced motion support, 44px minimum touch targets, and full RTL mirroring for Arabic.

PART 12

11. PWA versus Desktop Application

CorpusMind ships in two forms that share the same user interface, the same project file format, and the same analytical engine. The two forms differ in how the engine is delivered, where the data lives, and which platform features are available. This section documents the differences precisely so that researchers can choose the form that fits their workflow and their institutional data-governance constraints.

11.1 What the PWA Is

The Progressive Web App (PWA) is a browser-based version of CorpusMind hosted at <https://corpus-mind-web.vercel.app/>. It can be installed on Chrome, Edge, Safari, and Firefox by clicking the install icon in the address bar. Once installed, it runs in its own window, works offline after the first visit, and appears in the operating system's application list.

In the PWA configuration, the analytical engine runs either as a remote service (when the researcher has access to a self-hosted CorpusMind engine on a lab server) or as a process the researcher starts locally with the `corpusmind-engine` command and then connects to from the browser. The browser itself does not run Python; it talks to the engine over HTTP on `127.0.0.1:8765` or on a remote address the researcher configures.

11.2 What the Desktop Application Is

The desktop application is a Tauri 2 binary for Windows, macOS (Apple Silicon and Intel), and Linux. It bundles the web interface, the Rust supervisor process, and (when built with the sidecar spec) the PyInstaller- bundled engine into a single installable package. On launch, the Rust supervisor spawns the engine as a child process, polls its health endpoint until it is ready, redirects its stdout and stderr to log files, and kills it cleanly on application exit.

11.3 Feature Comparison Table

Capability	PWA	Desktop Application
Analytical tools (concordance, collocation, keyness, etc.)	Identical	Identical
Vision Suite (multimodal analysis)	Identical	Identical
AI Assistant (local LLM via Ollama)	Requires the researcher to run Ollama locally and configure the engine URL	The supervisor can spawn Ollama automatically if it is on the system PATH
AI Assistant (remote LLM via OpenAI / Anthropic)	Supported, opt-in	Supported, opt-in
Engine delivery	Researcher starts the engine manually, or connects to a lab-server engine	Supervisor spawns and supervises the engine automatically
At-rest encryption (AES-256-GCM)	Supported, with the encryption key set via environment variable	Supported, with the key set via the desktop settings dialog or the environment
File system access	Limited to files the researcher uploads through the browser dialog; no access to arbitrary paths on disk	Full access to files the researcher opens through the desktop dialog; can read from the project's data directory
Offline operation	After the first visit, the PWA shell loads from cache; the engine still needs to be running locally for analysis	Fully offline after installation; no network connection required for any analytical operation
Process lifecycle	The researcher starts and stops the engine manually; if the browser tab is closed, the engine keeps running until killed	The supervisor starts the engine on app launch and kills it on app exit; no orphaned processes
Log files	Engine logs go to wherever the researcher started the engine; the PWA does not manage them	Engine stdout and stderr are redirected to <code>engine.stdout.log</code> and <code>engine.stderr.log</code> in the OS log directory
Cross-platform consistency	Identical on every operating system because the browser is the runtime	Identical UI, but the supervisor and sidecar binary are platform-specific (one build per OS)
Installation	One click (install icon in the browser)	Download and run the platform installer (.dmg, .exe, .msi, .ApplImage, .deb)
Update mechanism	Transparent; the PWA updates on next visit when the server ships a new version	Manual; the researcher downloads the new installer or builds from source
Sandboxing	The browser sandbox restricts file system and network access to what the CSP allows	The Tauri shell enforces the same CSP, but the Rust supervisor has full process-control privileges
Institutional deployment	Easy: point the PWA at a lab-server engine URL and share the URL with the team	Each researcher installs the desktop app; a lab server is optional
Data residency	If the engine runs on a lab server, the corpus data lives on that server; if the engine runs locally, the data lives on the researcher's machine	The corpus data always lives on the researcher's machine; no data leaves the device unless a remote LLM provider is configured

11.4 Which Form to Choose

Choose the **PWA** if you want to try CorpusMind without installing anything, if your institution already hosts a shared engine on a lab server, or if you work across many machines and want your project state to follow you. The PWA is also the only option on ChromeOS and on locked-down corporate machines where you cannot install native applications.

Choose the **desktop application** if you work with sensitive or unpublished corpora and need a strict local-only default, if you want the engine to start and stop automatically with the application, if you

want the engine logs managed for you, or if you need the full offline experience with no browser in the loop. The desktop application is also the recommended form for classroom deployment where the instructor pre-installs the application on lab machines and students do not need to manage the engine lifecycle.

PART 13

12. CorpusMind Compared with Other Corpus Tools

CorpusMind is not a replacement for every existing corpus tool. Each tool in this section was designed for a different audience and a different era of computing. This section compares CorpusMind with four widely used corpus analysis tools: AntConc, Sketch Engine, #LancsBox, and Voyant Tools. The comparison is based on the official documentation of each tool as of 2025, and every claim is sourced to the tool's own website or to a peer-reviewed publication about it.

12.1 AntConc

AntConc is a freeware corpus analysis toolkit developed by Laurence Anthony at Waseda University and first released in 2002. It is a desktop application written in Perl and compiled to a native binary for Windows, macOS, and Linux. It is distributed at no cost from <https://www.laurenceanthony.net/software/antconc/>.

AntConc provides a concordancer (KWIC), a cluster and n-gram tool, a collocate tool, a word list tool, and a keyword tool. It supports regular-expression search, user-defined tag handling, and a range of output sort options. It does not include a lemmatiser, a POS tagger, or a dependency parser; the researcher is expected to pre-process the corpus with an external tool if lemmatised or POS-tagged search is required. It does not include an AI assistant, a vision pipeline, or multimodal analysis. Arabic support is limited because AntConc does not bundle an Arabic morphological analyser; researchers working with Arabic typically pre-process the corpus with a separate tool such as Farasa or CAMEL Tools and then import the tagged text into AntConc.

CorpusMind differs from AntConc in four respects. First, CorpusMind bundles spaCy and CAMEL Tools, so tokenisation, lemmatisation, POS tagging, dependency parsing, and Arabic morphology are available without any pre-processing step. Second, CorpusMind implements all seven collocation measures documented in the methodology literature (MI, T-score, log-likelihood, Dice, LogDice, chi-square, Delta P) and both keyness measures (log-likelihood ratio and Burrows's Zeta) in a single interface, whereas AntConc's collocate tool reports a single statistic selected by the user. Third, CorpusMind includes a Vision Suite for multimodal discourse analysis and an AI assistant with a citation-enforced contract, neither of which AntConc provides. Fourth, CorpusMind emits a YAML provenance record for every analytical operation, whereas AntConc does not record the pipeline configuration or the software version with the output. AntConc remains an excellent choice for teaching corpus linguistics at the introductory level, for quick concordance searches on

pre-processed corpora, and for researchers who prefer a single-purpose desktop tool without network dependencies.

12.2 Sketch Engine

Sketch Engine is a commercial corpus management and analysis platform developed by Lexical Computing Limited and first released in 2003. It is a web-based service available at <https://www.sketchengine.eu/>. It requires a paid subscription; academic personal accounts start at approximately 7.09 EUR per month, and institutional licences are priced per user. A free tier called SKELL is available for language learners but does not include the full analysis toolset.

Sketch Engine's signature feature is the Word Sketch: a one-page, automatically generated grammatical collocation profile of a word, built by parsing the corpus with a language-specific sketch grammar and grouping collocates by grammatical relation (subject-of, object-of, modifier-of, etc.). It also offers a distributional thesaurus, concordance, word list, keyword, n-grams, and term extraction. Sketch Engine hosts a large collection of ready-made corpora (British National Corpus, English Web Corpus 2021, enTenTen, and many others) and allows researchers to upload their own corpora up to a size limit determined by the subscription tier. It supports over 100 languages, with language-specific taggers and sketch grammars for most of them.

CorpusMind differs from Sketch Engine in three respects. First, CorpusMind is local-first and free (AGPL-3.0-only); the researcher's corpus never leaves their machine unless they explicitly configure a remote LLM provider. Sketch Engine is a cloud service; the corpus is uploaded to the Sketch Engine servers for processing. This makes Sketch Engine unsuitable for unpublished or confidential corpora unless the researcher has an institutional licence with a data-processing agreement. Second, CorpusMind does not currently implement the Word Sketch grammatical collocation profile; this is a feature on the roadmap for a later release. Sketch Engine's Word Sketch remains the reference implementation for grammatical collocation summarisation, and researchers who need it should use Sketch Engine. Third, CorpusMind includes a Vision Suite for multimodal discourse analysis and an AI assistant with a citation-enforced contract; Sketch Engine does not offer either. Sketch Engine is the better choice for researchers who need ready-made large reference corpora, for lexicographers who rely on Word Sketches, and for teams that prefer a managed cloud service over a local installation.

12.3 #LancsBox

#LancsBox is a free corpus analysis toolbox developed at Lancaster

University by Vaclav Brezina and colleagues and first released in 2015. The current version, #LancsBox X, is available at <https://lancsbox.lancs.ac.uk/>. It is a desktop application for Windows, macOS, and Linux and is distributed at no cost for non-commercial academic use.

#LancsBox provides concordancing, collocation analysis, frequency

lists, keyword analysis, dispersion plots, and a distinctive "Graph Coll" visualisation that shows collocational relationships as a network graph with the node word at the centre. It includes built-in support for BNC and BNC64 reference corpora and allows researchers to upload their own corpora.

It supports multiple languages for tokenisation but does not bundle language-specific POS taggers or morphological analysers beyond English.

CorpusMind differs from #LancsBox in three respects. First, CorpusMind implements the full seven-measure collocation suite and the four-measure dispersion suite (Juilland's D, Gries's DP, average reduced frequency, and average wait time) with provenance records, whereas #LancsBox focuses on a smaller set of measures with richer visualisation. Second, CorpusMind includes the Vision Suite and the AI assistant; #LancsBox does not. Third, CorpusMind bundles CAMEL Tools for Arabic morphology, dialect identification, and named-entity recognition, whereas #LancsBox treats Arabic as a plain-text corpus without morphological analysis.

#LancsBox is the better choice for researchers who value the Graph Coll

network visualisation, for classroom use where the visual immediacy of the tool supports teaching, and for researchers working primarily with English reference corpora.

12.4 Voyant Tools

Voyant Tools is a free, web-based text analysis environment developed by Stéfán Sinclair and Geoffrey Rockwell and first released in 2012. It is available at <https://voyant-tools.org/>. It is open source and can be self-hosted, but most researchers use the public hosted instance.

Voyant Tools provides a multi-panel interface with a reader, a word cloud, a trends graph, a contexts tool, a collocates tool, a correlations tool, a scatterplot, and a dreamcatcher visualisation. It is designed for digital-humanities work on small to medium corpora and emphasises visual exploration over statistical rigour. It does not bundle a POS tagger, a lemmatiser, or a dependency parser; it works on raw token frequency. It does not implement the standard collocation measures (MI, t-score, log-likelihood, etc.) and does not emit a provenance record. It does not support Arabic morphology.

CorpusMind differs from Voyant Tools in four respects. First, CorpusMind implements the full statistical suite (seven collocation measures, four dispersion measures, two keyness measures) with formal definitions and primary-literature citations, whereas Voyant Tools focuses on visual exploration. Second, CorpusMind bundles spaCy and CAMEL Tools for linguistic annotation, whereas Voyant Tools works on raw tokens. Third, CorpusMind includes the Vision Suite and the AI assistant; Voyant Tools does not. Fourth, CorpusMind emits a YAML provenance record for every operation; Voyant Tools does not. Voyant Tools is the better choice for digital-humanities researchers who want a quick visual overview of a text, for teaching introductory text analysis, and for researchers who do not need linguistic annotation or statistical rigour.

12.5 Summary Comparison Table

Feature	CorpusMind	AntConc	Sketch Engine	#LancsBox	Voyant Tools
Licence	AGPL-3.0-only (free, open source)	Freeware (closed source)	Commercial (subscription)	Free for academic use (closed source)	Open source (GPL)
Price	Free	Free	From 7.09 EUR/month (academic)	Free	Free
Deployment	Local-first desktop + PWA	Desktop only	Cloud only	Desktop only	Web only
Concordance (KWIC)	Yes	Yes	Yes	Yes	Yes (Contexts)
Collocation measures	7 (MI, t-score, LL, Dice, LogDice, chi-square, Delta P)	1 (user-selected)	Multiple (Word Sketch grammar-based)	Several	1 (raw co-occurrence)
Keyness	Yes (LL, chi-square, Log Ratio, %DIFF, Simple Maths, Odds Ratio)	Yes (LL, chi-square)	Yes	Yes	No
Dispersion	Yes (Juilland's D, Gries's DP, ARF, AWT)	No	Yes	Yes (visual)	No
POS tagging	Yes (spaCy)	No (pre-process externally)	Yes (language-specific)	Limited	No
Lemmatisation	Yes (spaCy, CAMEL Tools)	No	Yes	Limited	No
Dependency parsing	Yes (spaCy)	No	No	No	No
Arabic morphology	Yes (CAMEL Tools: root, pattern, lemma, dialect ID)	No	Yes (Arabic sketch grammar)	No	No
Multimodal / Vision analysis	Yes (Kress and van Leeuwen Visual Grammar, OCR, object detection)	No	No	No	No
AI assistant	Yes (grounded, citation-enforced, local LLM via Ollama)	No	No	No	No
Provenance records (YAML)	Yes (every operation)	No	No	No	No
At-rest encryption	Yes (AES-256-GCM)	No	No (cloud-hosted)	No	No (cloud-hosted)
Ready-made reference corpora	No (researcher supplies corpus)	No	Yes (BNC, enTenTen, 100+ languages)	Yes (BNC, BNC64)	No
Word Sketch (grammatical collocation profile)	No (on roadmap)	No	Yes (reference implementation)	No	No
Network collocation visualisation	No (table-based)	No	No	Yes (Graph Coll)	No
Word cloud	No	No	No	No	Yes
Offline operation	Yes (desktop) / partial (PWA)	Yes	No (cloud)	Yes	No (web)
Data residency	Local by default	Local	Cloud (uploaded)	Local	Cloud (uploaded, or self-host)

Feature	CorpusMind	AntConc	Sketch Engine	#LancsBox	Voyant Tools
Reproducibility for peer review	Yes (YAML provenance + methods.pdf export)	Manual	Limited	Manual	Manual

12.6 When to Use Which Tool

Use **CorpusMind** when you need local-first, reproducible, framework-grounded analysis of English or Arabic corpora, when you need multimodal discourse analysis of images alongside text, or when you need an AI assistant that grounds its claims in corpus evidence.

Use **AntConc** when you need a quick concordance on a pre-processed corpus in a classroom or workshop setting, when you do not need lemmatisation or POS tagging, or when you are on a machine where you cannot install Python.

Use **Sketch Engine** when you need ready-made large reference corpora, when you need the Word Sketch grammatical collocation profile, or when your team prefers a managed cloud service and your data-governance policy permits uploading the corpus to a third-party server.

Use **#LancsBox** when you value the Graph Coll network visualisation, when you are teaching corpus linguistics and want a visually intuitive tool, or when you work primarily with the BNC or BNC64.

Use **Voyant Tools** when you want a quick visual overview of a text in a digital-humanities context, when you do not need linguistic annotation or statistical rigour, or when you are working in a browser-only environment.

These tools are complementary, not mutually exclusive. Many researchers will use more than one: for example, Voyant Tools for an initial visual overview, CorpusMind for the rigorous statistical and multimodal analysis, and Sketch Engine for the Word Sketch when a grammatical collocation profile is required.

PART 14

Getting Help

- bullet GitHub: <https://github.com/waleedmandour/CorpusMind/issues>
- bullet Live PWA: <https://corpus-mind-web.vercel.app/>
- bullet Build Guide: docs/BUILD_GUIDE.md
- bullet Methodology Reference: docs/METHODOLOGY.md
- bullet Zenodo DOI: <https://doi.org/10.5281/zenodo.21226650>